Semantic Consistency Matters: Language-Driven Trigger-Aligned Diffusion for Multimodal Remote Sensing Model Watermarking

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***Abstract*—Deep learning has significantly advanced the development of multimodal remote sensing, playing a crucial role in various Earth observation tasks. Due to the high cost of multimodal data annotation and computational resources, well-performing multimodal remote sensing image classification models are a valuable asset, rendering the models vulnerable to malicious attacks. Model watermarking has emerged as an effective solution for intellectual property (IP) protection of deep learning models. However, most existing model watermarking methods are tailored for single-modal models and follow a modality-independent paradigm that ignores cross-modal interactions, thus limiting the triggering effectiveness of the model watermark. To solve this problem, we propose a language-driven trigger-aligned diffusion method (LTDifMark) for multimodal remote sensing image classification model watermarking, enabling reliable IP protection. LTDifMark aims to generate semantic consistent multimodal trigger sample with the guidance of multimodal shared features, thereby ensuring the stealthiness and collaborative triggering capability of the model watermark. In particular, the language-guided multimodal feature alignment strategy is designed to utilize class-specific language priors to align multimodal features in the same space. The proposed shared feature-conditioned Mamba-diffusion trigger sample generation method leverages extracted multimodal shared features to collaboratively generate multimodal trigger samples, balancing the effectiveness and stealthiness of trigger sets. Experiments demonstrate that the proposed method achieves superior effectiveness compared with state-of-the-art methods and exhibits robustness against various attacks.**

***Index Terms*—Diffusion model, language-driven, model watermarking, multimodal remote sensing image.**

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I. Introduction

DEEP neural networks (DNNs) have achieved remarkable progress in various remote sensing tasks, including classification [1], [2], [3], semantic segmentation [4], and change detection [5]. A multimodal remote sensing image classification model analyzes land-cover information with different characteristics of ground objects [6], [7], [8], which is essential for environmental monitoring and resource management. However, acquiring multimodal images requires expensive imaging sensors, and manual label annotation is time-consuming, making massive labeled data difficult to obtain [9], [10], [11], [12]. Moreover, due to the large-scale remote sensing data, model training requires substantial computational resources [13], [14]. Therefore, a multimodal remote sensing image classification model with high performance is an expensive digital asset, which may be unauthorized duplicated and distributed by adversaries [15], [16]. It is an urgent need to protect the intellectual property (IP) of multimodal remote sensing image classification models, ensuring the reliability of models in practical applications.

Digital watermarking technology has been applied to embed an imperceptible digital watermark into multimedia data for copyright protection. However, it is infeasible to be directly applied to DNNs due to potential performance degradation. Therefore, DNN watermarking approaches are developed to verify the owner of models, thus protecting the IP of DNN models. Based on the requirement of model details in extraction, DNN watermarking methods can be categorized into two categories: white-box and black-box DNN watermarking [17]. In white-box DNN watermarking, the owner embeds watermarks in the weights or architecture of DNN models, typically realized by designing a loss term for optimization [18], [19]. These methods require the suspicious model to expose the internal details to the owner, which is hard to implement in real-world scenarios. Black-box DNN watermarking methods assume that only the API is available and ownership is verified by interacting with the target model, without requiring internal

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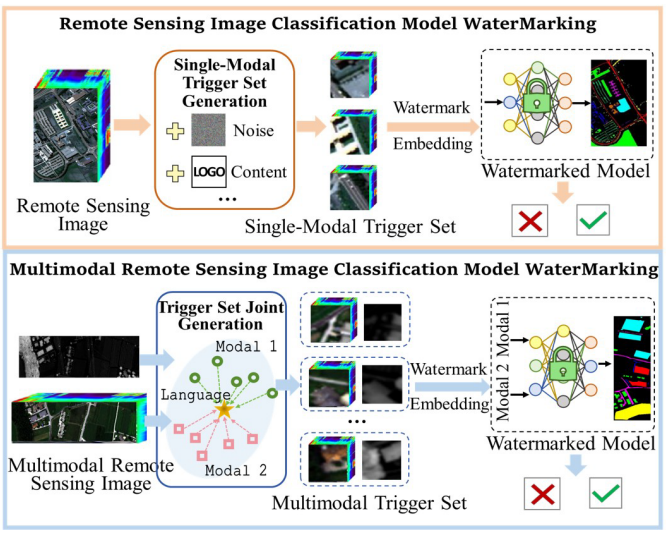


Fig. 1. Difference between conventional single-modal and proposed multimodal remote sensing image classification model watermarking methods.

parameters or architecture of the DNN [20], [21]. They force the DNN to fit a carefully designed trigger set and verify ownership by comparing the outputs of watermarked and suspicious models on trigger samples.

However, as shown in Fig. 1, most of the existing DNN watermarking methods are tailored to protect single-modal classification models. Directly extending to multimodal classification models typically involves processing each modality independently in parallel, treating each modality as a separate information stream. The modality-independent paradigm neglects the inherent collaborative relationship across modalities, severely hindering the model from learning specific mappings of trigger samples as watermarking. This naturally raises a question: *how to leverage the cross-modal dependencies to improve the multimodal triggering capability of model watermarking?*

To solve this problem, we propose a language-driven trigger-aligned diffusion method (LTDifMark) for multimodal remote sensing image classification model watermarking, which extracts class-specific shared features to ensure semantically consistent multimodal trigger sample generation, thereby enhancing the stealthiness of the trigger set and the collaborative triggering capability of the model watermark. In particular, the language-guided multimodal collaborative trigger set generation method is designed to utilize language priors to drive multimodal shared feature alignment and leverage the extracted multimodal shared features to guide the Mamba-diffusion model to construct a semantically consistent multimodal trigger set. The multimodal trigger set is encouraged to be classified as the predefined label for watermark embedding. For ownership verification, the model owner queries the suspicious model with the multimodal trigger samples and verifies consistency with the predefined trigger-label mappings. The main contributions of this article are summarized as follows.

1) We propose a language-driven trigger-aligned diffusion method (LTDifMark) for multimodal remote sensing model watermarking, which exploits the collaborative dependencies of multimodal remote sensing data to enable triggering capability improvement of the model watermark, providing the first dedicated solution to protect the IP of multimodal remote sensing image classification models.
2) We design a language-guided multimodal feature alignment strategy, which introduces language priors to explicitly guide class-specific shared feature learning of both modalities in the same space.
3) We propose a shared feature conditioned Mamba-diffusion trigger sample generation method that leverages multimodal shared features to drive the Mamba-diffusion model to generate multimodal trigger samples collaboratively, thus guaranteeing semantic consistency and stealthiness for reliable watermark embedding.

The remainder of this article is structured as follows. Section II provides the related work. Section III describes the details of the proposed method. Section IV evaluates the performance on various models. Finally, Section V concludes this article.

II. Related Work

A. DNN Watermarking

In the field of remote sensing, the rapid evolution of deep learning has led to the emergence of high-value models with sophisticated capabilities [22], [23]. For instance, to handle diverse real-world challenges, such as cross-scene distribution shifts and the continuous emergence of novel categories [24], various advanced learning paradigms have been proposed recently. To tackle the domain shift problem across different scenes, parameter-efficient foundation models like SpectralX [25] have been developed for spectral domain generalization. Similarly, bidirectional domain adaptation [26] and single-source domain expansion networks [27] are presented to achieve robust cross-domain classification. To continuously learn new land-cover categories, class-incremental learning methods [28] have been widely explored to overcome the catastrophic forgetting of previous knowledge. Meanwhile, long-tailed classification methods [29] have also been developed to improve the recognition reliability of minority land-cover categories under uneven sample distributions. As these models become increasingly essential and expensive digital assets due to their high generalization and technical complexity, protecting their IP has become an urgent necessity. DNN watermarking aims to embed a series of unique information into the model, thus identifying the owner of the model. DNN watermarking is an effective IP protection technique to determine whether a suspicious model is an unauthorized replica by a third party. DNN watermarking methods are divided into two classes based on the model details accessible to the ownership verifier: white-box DNN watermarking and black-box DNN watermarking.

White-box DNN watermarking methods embed designated information, such as bit strings, directly in the model parameters as watermarks. In the white-box scenario, the weight parameters and activation maps of the model are publicly

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accessible. Darvish Rouhani et al. [30] embedded the watermark string into the low-probability regions of different layers of the model. DeepMarks [31] encoded the collusion-aware fingerprint of users in the weights of the model, enabling both ownership verification and user tracking. Fan et al. [32] and Zhang et al. [33] adopted passport layers to carry a watermark, which effectively resists network modification and ambiguity attacks. HuFuNet [34] embedded the encoder into the model, and the owner used the decoder for infringement detection, which theoretically guarantees robustness against fine-tuning attacks. Li et al. [35] proposed an effective and harmless model ownership verification method, which extracts the knowledge of defender-specified external features to verify the ownership. However, model stealers are likely to refuse to provide internal model details, making the white-box verification difficult to implement in reality.

Black-box DNN watermarking methods enable ownership verification without requiring detailed knowledge of the model, which is more practical in real-world scenarios [36]. Zhang et al. [37] generated trigger samples by three different strategies, obtaining content-based, unrelated-based, and noise-based watermarks. Li et al. [38] embedded external features from training samples with style transfer, which adopts defender-specific information to defend against model stealing. SSW [39] adopted a pair of symmetric shadow models to select the trigger set progressively. Liu et al. [40] added low-intensity trigger perturbations to the sensitive frequencies of clean samples, thus improving the imperceptibility of trigger samples. MEA-Defender [41] combined samples from two classes to obtain watermark samples, which are essential in the training set to protect DNN models against model extraction. Note that the black-box DNN watermarking methods are designed to verify the ownership of single-modal classification models. In this article, we explore the collaborative relationship between multimodal images, enhancing the capability to protect the IP of multimodal remote sensing image classification models.

B. Diffusion Models

Recently, diffusion models [42] have shown strong capability for image generation. A diffusion model consists of a forward noising process and a reverse denoising process. Gaussian noise is gradually added to the data in the forward process, and a U-Net is constructed to predict the noise in each step in the reverse process to obtain the desired image. Denoising diffusion probabilistic model (DDPM) is a discretized version of a diffusion model, which uses parameterized Markov chains to generate samples after a finite amount of time. Compared to generative adversarial networks (GANs), diffusion models are more stable in the training process and have been widely used in various computer vision fields such as super-resolution [43] and adversarial attack. For example, Chen et al. [44] add perturbations in the latent space of diffusion models to embed human-insensitive perturbations into semantic clues. Advdiffuser [45] is proposed to generate natural unrestricted adversarial examples, which learns class activation mapping to perturb less important areas. Diff-PGD [46] exploits a gradient guided by a diffusion model to generate adversarial samples with higher stealthiness. These studies demonstrate that diffusion models can generate adversarial samples for single-modal classification models, which introduce imperceptible perturbations while preserving semantic consistency. Inspired by this capability, we aim to investigate the potential for generating imperceptible multimodal trigger samples.

C. State Space Models

State space models (SSMs) are adept at modeling sequential dependencies, which were initially proposed for natural language processing. Mamba [47] is a novel type of SSM that models long-distance dependence relationships while maintaining linear complexity, which has gained popularity in various computer vision domains, such as image classification and change detection [48]. VMamba [49] is further proposed as a vision backbone, which designs a 2-D selective scan for spatial domain traversal, facilitating efficient visual representation learning. Vim [50] marks the image sequences with position embeddings and designs bidirectional SSM for global visual context modeling. In this article, Mamba is extended to the trigger sample generation domain, mining the capability of SSMs to enhance the performance of remote sensing image generative tasks.

III. Proposed Method

A. Threat Model

In this section, we formulate the threat model to characterize the goals and capabilities of the model owner and adversary in the black-box setting. Since DNNs are of great value, there is a risk that an adversary might illegally duplicate or misappropriate the model for their own advantage. The proposed watermarking method aims to embed ownership information into a multimodal joint classification model to verify the ownership of a suspicious model. Therefore, multimodal DNN watermarking methods are expected to balance fidelity and robustness, which minimizes the effect on the performance of multimodal classification models while exhibiting robustness against attacks. It is worth noting that the protected object of the proposed method is not a set of independently deployed single-modality classification models. Although the multimodal model may contain modality-specific branches, these branches are internal components of the complete joint model and are not independently watermarked as standalone models. Therefore, accessing only one modality-specific branch or using ordinary single-modality data does not correspond to the ownership verification scenario considered in this work.

1) *Model Owner:* The model owner invests a large amount of effort and resources to train a well-performing multimodal remote sensing image classification model. To protect the IP of the well-performing model, the owner implants a watermark into the model before providing access to users via APIs. When there exists unauthorized use by other parties, the owner has the ability to verify ownership.
    a) The model owner, who has sufficient computational resources, possesses comprehensive knowledge of

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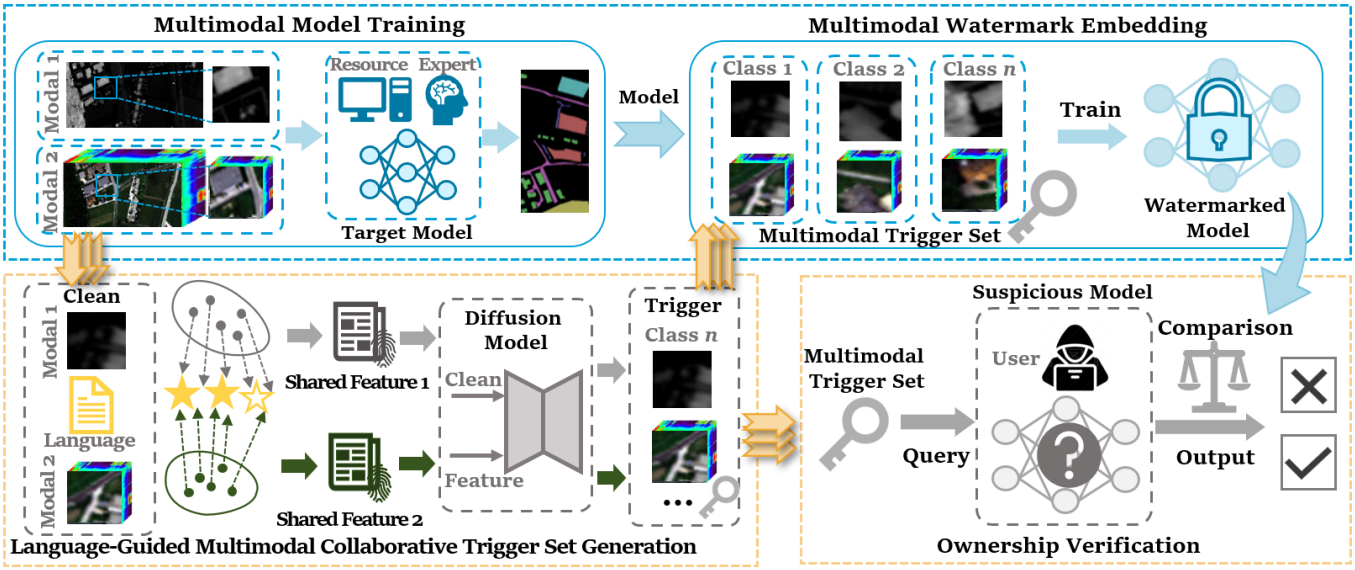


Fig. 2. Framework of a language-driven trigger-aligned diffusion method (LTDifMark) for multimodal remote sensing image classification model watermarking.

the detailed information of DNNs, such as the model architecture, parameters, and training set.

b) When infringement is detected, the model owner only has access to the suspicious model via API, which is a black-box scenario. For ownership verification, the model owner queries the suspicious complete multimodal model with the privately preserved paired HSI-LiDAR trigger samples. Ordinary HSI-only or LiDAR-only samples do not satisfy the proposed verification protocol, because they are neither paired multimodal trigger samples nor associated with the owner-defined trigger-label mappings.

2) *Adversary:* The adversary may steal the model through data leakage of the owner, unencrypted storage or transmission, or abuse of model access permissions, and then deliberately claim the stolen model as self-developed.

a) The adversary can remove the watermark from the victim model by using various attacks, such as fine-tuning, pruning, and model distillation.

b) The adversary lacks sufficient training data and computational resources, making it impractical to independently train a high-performing model from scratch.

B. Overview

Given a well-performed multimodal remote sensing image classification model as target model $M_{\mathcal{O}}$, the proposed model watermarking method aims to embed ownership information into $M_{\mathcal{O}}$ to obtain watermarked model $M_{\mathcal{O}}^{\mathcal{W}}$, thereby protecting IP of target model from unauthorized copying, misuse or theft. Most existing methods are designed for single-modal classification models, but these methods generate trigger samples of each modality separately and ignore the mutual interference between them when applied directly to multimodal classification model protection. To address this problem, as shown in Fig. 2, LTDifMark is proposed for multimodal remote sensing model IP protection, which involves three consecutive phases, i.e., language-guided multimodal collaborative trigger set generation, multimodal watermark embedding, and ownership verification. The core idea of the proposed method is to explore the collaborative trigger ability of different modalities for IP protection of a multimodal classification model while ensuring the imperceptibility of the generated multimodal trigger samples.

1) Language-Guided Multimodal Collaborative Trigger Set Generation: Under the explicit guidance of language priors of each class $\mathcal{T}_k = \{\mathbf{T}_k\}_{k=1}^{K}$, the multimodal clean samples $\mathcal{C} = \{\mathbf{X}_c^{v_1}, \mathbf{X}_c^{v_2}, y_c\}_{c=1}^{C}$ are aligned to the same space to obtain class-specific share features of each modality, where K and C are the number of classes and clean samples, respectively. The extracted share features are adopted as condition of Mamba-diffusion network to derive multimodal clean samples modification, generating multimodal trigger samples with satisfactory imperceptibility $\mathcal{A} = \{(\widehat{\mathbf{X}}_a^{v_1}, \widehat{\mathbf{X}}_a^{v_2}, y_a)\}_{a=1}^{A}, (A = C)$. The multimodal trigger set generation process is formulated as follows:

$$\left(\widehat{\mathbf{X}}_a^{v_1}, \widehat{\mathbf{X}}_a^{v_2}\right) = \mathcal{F}_{Tr}\Big[\mathcal{F}_{Ma}\left[\left(\mathbf{X}_c^{v_1}, \mathbf{X}_c^{v_2}, y_c\right), \mathbf{T}_c\right] \left(\mathbf{X}_a^{v_1}, \mathbf{X}_a^{v_2}, y_a\right)\Big], \quad y_c \neq y_a \tag{1}$$

where $\mathcal{F}_{Ma}(\cdot)$ and $\mathcal{F}_{Tr}(\cdot)$ denote language-guided multimodal feature alignment and shared feature-conditioned Mamba-diffusion trigger sample generation functions, respectively.

2) Multimodal Watermark Embedding: The owner embeds the model watermarking by training the target model $M_{\mathcal{O}}$ with both the clean training set and multimodal trigger samples $\mathcal{A}$, thereby modifying the parameters θ_0 as θ^*. We assign a designated label y_p to the multimodal trigger samples, and the multimodal watermark embedding can be formalized as

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follows:

$$\theta^* \leftarrow \arg\min_{\theta_0} \sum_{j=1}^{N} \mathcal{L}\left(M_{\mathcal{O}}\left(\theta_0; \widehat{\mathbf{X}}_a^{v_1}, \widehat{\mathbf{X}}_a^{v_2}\right), y_p\right). \tag{2}$$

3) Ownership Verification: To verify the ownership of suspicious model $M_{\mathcal{SP}}(\theta')$, the owner queries the model with the embedded multimodal trigger samples and compares the outputs with the predefined labels

$$Verify\left(M_{\mathcal{SP}}\left(\theta'; \widehat{\mathbf{X}}_a^{v_1}, \widehat{\mathbf{X}}_a^{v_2}, y_p\right)\right) = \begin{cases} \text{True}, & \text{p-value} < \alpha \\ \text{False}, & \text{otherwise} \end{cases}. \tag{3}$$

Algorithm 1 Overall Watermarking Process of LDTifMark

Input: Clean multimodal training set $\mathcal{C} = \{\mathbf{X}_c^{v_1}, \mathbf{X}_c^{v_2}, y_c\}_{c=1}^{C}$; target model $M_{\mathcal{O}}$; text encoder $TextEnc(\cdot)$; watermark label y_c; verification threshold α.
Output: Watermarked model $M_{\mathcal{O}}^{\mathcal{W}}$; multimodal trigger set $\mathcal{T}_M$; ownership verification result.

Language-guided multimodal feature alignment;
for *each class* k **do**
  Construct the language description $\mathbf{T}_k$ using class name, common attributes, and inter-class correlations;
  Extract the language feature $\mathbf{T}'_k = TextEnc(\mathbf{T}_k)$;
Extract modality-specific visual features $\mathbf{F}^{v_1}$ and $\mathbf{F}^{v_1}$ under the language-visual alignment constraint;
Multimodal collaborative trigger set generation;
Select clean multimodal samples as source samples;
Feed the source samples and aligned shared features into the shared-feature conditioned Mamba-diffusion model;
Generate multimodal trigger pairs $(\widehat{\mathbf{X}}_a^{v_1}, \widehat{\mathbf{X}}_a^{v_2})$ through the reverse diffusion process by optimizing the trigger samples using the classification loss and visual consistency loss;
Watermark embedding;
Insert $(\widehat{\mathbf{X}}_a^{v_1}, \widehat{\mathbf{X}}_a^{v_2})$ into the training phase of the target model;
Fine-tune $M_{\mathcal{O}}$ using both clean samples and multimodal trigger samples;
Obtain the watermarked model $M_{\mathcal{O}}^{\mathcal{W}}$;
Black-box ownership verification;
Query the suspicious model M_{SP} with $(\widehat{\mathbf{X}}_a^{v_1}, \widehat{\mathbf{X}}_a^{v_2})$ through the API;
Obtain the predicted labels and calculate the trigger-set accuracy;
if *the trigger-set accuracy is higher than* α **then**
  Identify $M_{\mathcal{SP}}$ as an unauthorized copy of $M_{\mathcal{O}}^{\mathcal{W}}$;
else
  Regard $M_{\mathcal{SP}}$ as an independent model;

The overall implementation flow of the proposed LDTifMark is summarized in Algorithm 1.

C. Language-Guided Multimodal Collaborative Trigger Set Generation

1) Language-Guided Multimodal Feature Alignment: For multimodal remote sensing watermarking, trigger sample generation requires not only visual imperceptibility within each modality, but also cross-modal semantic consistency between different modalities. Natural RGB image watermarking mainly emphasizes visual imperceptibility in RGB appearance, whereas HSI and LiDAR data contain spectral and structural information that is closely related to land-cover characteristics. Due to complex ground-object distributions and heterogeneous background patterns, visually slight perturbations may still introduce abnormal spectral or structural responses, or weaken the consistency between modalities. Therefore, multimodal trigger samples should not only be inconspicuous in each modality, but also maintain reasonable cross-modal semantic consistency.

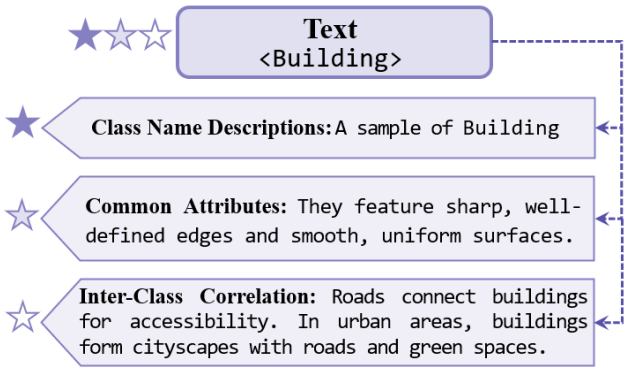


Fig. 3. Examples of language descriptions.

Since a class name, such as building or trees, only provides a coarse category-level cognition and cannot comprehensively describe land-cover characteristics across different modalities, language descriptions are introduced to provide more explicit semantic guidance for multimodal feature alignment. Existing CLIP-style vision-language alignment and cross-modal image-text retrieval methods mainly learn visual-textual correspondence for image-text association [51]. Directly applying this paradigm may separately align multimodal features with text, which fails to explicitly capture the common semantics required by both modalities. To address this problem, class-level language descriptions are constructed that contain class names, common attributes, and interclass correlations. These descriptions encode land-cover knowledge shared by multimodal data, rather than instance-specific captions for a single image. Therefore, they serve as modality-shared semantic anchors to guide class-specific shared feature learning, providing consistent semantic conditions for subsequent Mamba-diffusion trigger generation.

In particular, the language description of each class is constructed from three aspects, including class name description, common attributes, and interclass correlations, which jointly provide category-level, intraclass, and interclass semantic guidance for learning class-specific shared features. The class name provides category-level semantic guidance, common attributes describe intraclass modality-shared characteristics, and interclass correlations introduce contextual priors among different land-cover categories. For different datasets, the same construction principle is maintained, while the specific descriptions are instantiated according to the corresponding land-cover categories and scene characteristics. As shown in Fig. 3, class name description follows a fixed template "`a sample of [cls]`" to expand the class description. Common attributes $\{\text{ComAttri}\}_k$ include the texture characteristics, shape of the ground object, and other common information

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that is consistent across different modalities. Interclass correlation $\{\text{InterCorr}\}_k$ refers to the potential associations or spatial arrangements between different classes. The language descriptions $\mathbf{T}_k$ of class k is formed as follows:

$$\mathbf{T}_k = \left\{\text{A sample of [cls]}\right\}_k, \left\{\text{ComAttri}\right\}_k, \left\{\text{InterCorr}\right\}_k. \quad (4)$$

As shown in Fig. 4, the construed language description is encoded to obtain class-specific language features by adopting the pretrained transformer architecture of CLIP. The lower cased byte pair encoding (BPE) is utilized to generate tokenized representations, and then fed to the transformer with fixed parameters to extract the class-specific language features $\mathbf{T}'_k$

$$\mathbf{T}'_k = TextEnc\left(\mathbf{T}_k\right) \quad (5)$$

where $TextEnc(\cdot)$ represents the function of text encoder.

Due to its capability in modeling long-distance dependence relationships and maintaining linear computational complexity, Mamba has been introduced in computer vision tasks and achieved superior performance than Transformer. The visual encoder, consisting of VSS blocks, is designed to extract multimodal features of remote sensing images. The visual encoder takes advantage of VSS, which provides a comprehensive understanding of spatial and spectral information, thus enhancing the ability to extract global features. For modality v_1, the visual feature $\mathbf{F}_c^{v_1}$ is formulated as follows:

$$\begin{aligned}\mathbf{F}_c^{v_1} &= VisEnc\left(\mathbf{X}_c^{v_1}\right)\\ &= VSS^L\left(\cdots VSS^l\left(\cdots VSS^1\left(\mathbf{X}_c^{v_1}\right)\right)\right)\end{aligned} \quad (6)$$

where $VisEnc(\cdot)$ is the visual encoder of modality v_1 and $VSS^l(\cdot)$ denotes the lth VSS blocks. The $l-1$th visual features of modality $v_1\mathbf{F}_c^{v_1,l-1}(\mathbf{F}_c^{v_1,1} = \mathbf{X}_c^{v_1})$ are first fed into two parallel linear layers f^{Linear}, where one branch incorporates a SiLU activation f^{SiLU} to generate gating signals $\mathbf{F}_1^{v_1,l}$

$$\mathbf{F}_{c,1}^{v_1,l} = f^{SiLU}\left(f^{Linear}\left(\mathbf{F}_c^{v_1,l-1}\right)\right). \quad (7)$$

For the second branch, the features are then passed through depthwise convolution f^{DWConv}, 2-D-selective-scan (SS2D), SiLU activation, and layer norm f^{LN} to obtain output feature $\mathbf{F}_{c,2}^{v_1,l}$. With SS2D as the core computing unit, the VSS block selectively scans the image patches in four directions and combines these resulting features to capture long-range dependencies. The output feature $\mathbf{F}_{c,2}^{v_1,l}$ of the second branch is defined as follows:

$$\mathbf{F}_{c,2}^{v_1,l} = f^{LN}\left(SS2D\left(f^{SiLU}\left(f^{DWConv}\left(f^{Linear}\left(\mathbf{F}_c^{v_1,l-1}\right)\right)\right)\right)\right). \quad (8)$$

Finally, the output feature of the lth VSS block is generated by fusing the features of both branches, followed by a linear layer

$$\mathbf{F}_c^{v_1,l} = f^{Linear}\left(\mathbf{F}_{c,1}^{v_1,l} \otimes \mathbf{F}_{c,2}^{v_1,l}\right) + \mathbf{F}_c^{v_1,l-1} \quad (9)$$

where $\otimes$ represents the Hadamard product. Similarly, the visual features of modality $v_2\mathbf{F}_c^{v_2}$ are extracted through a visual encoder consisting of stacked VSS blocks, with the encoder architecture kept consistent across modalities.

Given language features $\mathbf{T}'$ and multimodal visual features $\mathbf{F}^{v_1}$ and $\mathbf{F}^{v_2}$, the multimodal visual-language alignment loss $\mathcal{L}_{align}$ is designed to guide the visual encoder to focus on learning class-specific shared features of each modality in the same space. The proposed loss applies contrastive learning to each modality, pulling multimodal visual features closer to the corresponding class-specific language features while pushing away features of different classes

$$\mathcal{L}_{align} = \mathcal{L}_{con}\left(\mathbf{F}^{v_1}, \mathbf{T}'\right) + \mathcal{L}_{con}\left(\mathbf{F}^{v_2}, \mathbf{T}'\right) \quad (10)$$

where $\mathcal{L}_{con}(\cdot)$ is defined for both language-to-visual alignment and visual-to-language alignment

$$\begin{aligned}&\mathcal{L}_{con}\left(\mathbf{F}^{v_m}, \mathbf{T}'\right)\\ &= -\sum_{n=0}^{N}\frac{1}{|P(n)|}\Bigg(\sum_{p\in P_v(n)}\log\frac{\exp\left(\mathbf{T}_n'^{\mathrm{T}}\mathbf{F}_p^{v_m+}/\tau\right)}{\sum_{a\in A_v(n)}\exp\left(\mathbf{T}_n'^{\mathrm{T}}\mathbf{F}_a^{v_m-}/\tau\right)}\\ &\qquad+\sum_{p\in P_l(n)}\log\frac{\exp\left(\mathbf{F}_n^{v_m\mathrm{T}}\mathbf{T}_p'^{+}/\tau\right)}{\sum_{a\in A_l(n)}\exp\left(\mathbf{F}_n^{v_m\mathrm{T}}\mathbf{T}_a'^{-}/\tau\right)}\Bigg)\end{aligned} \quad (11)$$

where $\mathbf{F}^{v_m}$ denotes the feature of modalities v_m in minibatch, P_v and A_v are positive and negative visual sample set for the nth language sample, P_l and A_l are positive and negative visual sample set for the nth visual sample, $\mathbf{F}_p^{v_m+}$ and $\mathbf{F}_a^{v_m-}$ are positive and negative visual samples, $\mathbf{T}_p'^{+}$ and $\mathbf{T}_a'^{-}$ are positive and negative language samples, $|\cdot|$ calculates the number of samples, and τ represents a temperature parameter.

2) Shared Feature Conditioned Mamba-Diffusion Trigger Sample Generation: Compared to the trigger samples for single-modal image classification model, multimodal trigger sample generation involves adding information to images of both modalities, requiring the consideration of two key aspects: a) the trigger samples of different modalities should collaborate effectively to enhance the trigger ability, without weakening the effectiveness of either modality; and b) the generated multimodal trigger sample should maintain the realism of the clean sample, remaining imperceptible to humans to enhance the stealthiness of the model watermarking method. Therefore, the shared feature conditioned Mamba-diffusion trigger sample generation method is proposed to obtain multimodal trigger sample pairs $(\widehat{\mathbf{X}}_a^{v_1}, \widehat{\mathbf{X}}_a^{v_2})$, which leverages the class-specific shared feature of each modality in the same space as a condition to drive the proposed Mamba-diffusion model, balancing between the collaborative trigger ability and imperceptibility. The shared features are obtained from the language-guided multimodal feature alignment module and encode consistent target-class semantics across modalities. At each reverse diffusion step, the noisy sample of each modality is concatenated with its corresponding shared feature and then fed into the Mamba-based denoising network to predict the noise component. Since both modality branches are conditioned on semantically aligned shared features and supervised by the same target label, their denoising trajectories are guided toward consistent class semantics while retaining their modality-specific visual characteristics. In this way, the generated multimodal trigger pairs remain semantically aligned rather than being independently optimized for each modality. The objective of the proposed multimodal trigger

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sample generation method is defined as follows:

$$\begin{cases} \widehat{\mathbf{X}}_a^{v_1} = MamDif\left(\mathbf{X}_a^{v_1}, \mathbf{F}_c^{v_1}, y_a, y_c\right) \\ \widehat{\mathbf{X}}_a^{v_2} = MamDif\left(\mathbf{X}_a^{v_2}, \mathbf{F}_c^{v_2}, y_a, y_c\right) \\ \left(\widehat{\mathbf{X}}_a^{v_1}, \widehat{\mathbf{X}}_a^{v_2}\right) = \underset{\left(\widehat{\mathbf{X}}_a^{v_1}, \widehat{\mathbf{X}}_a^{v_2}\right)}{\arg\min} \mathcal{L}_{CE}\left(M_{\mathcal{O}}\left(\widehat{\mathbf{X}}_a^{v_1}, \widehat{\mathbf{X}}_a^{v_2}\right), y_a\right) \end{cases}$$
$$\text{s.t. } \left\|\widehat{\mathbf{X}}_a^{v_1} - \mathbf{X}_a^{v_1}\right\|_2 \le \epsilon \text{ and } \left\|\widehat{\mathbf{X}}_a^{v_2} - \mathbf{X}_a^{v_2}\right\|_2 \le \epsilon \tag{12}$$

where $MamDif(\cdot)$ denotes the function of the shared feature-conditioned Mamba-diffusion model.

The proposed shared feature-conditioned Mamba-diffusion model consists of a forward process and a reverse process to generate trigger samples of each modality, and the DDPM sampling strategy is adopted to improve quality. The forward process of DDPM defines a Markov chain and gradually adds Gaussian noise to corrupt the original data $\mathbf{X}_{a,0}^{v_m}$

$$q_\sigma\left(\mathbf{X}_{a,1:T}^{v_m} \mid \mathbf{X}_{a,0}^{v_m}\right) := \prod_{t=1}^{T} q_\sigma\left(\mathbf{X}_{a,t}^{v_m} \mid \mathbf{X}_{a,t-1}^{v_m}\right). \tag{13}$$

The conditioned probability $q_\sigma(\mathbf{X}_{a,t}^{v_m} \mid \mathbf{X}_{a,t-1}^{v_m})$ is formulated as a Gaussian distribution

$$q_\sigma\left(\mathbf{X}_{a,t}^{v_m} \mid \mathbf{X}_{a,t-1}^{v_m}\right) = \mathcal{N}\left(\mathbf{X}_{a,t}^{v_m};\ \sqrt{1-\beta_t}\mathbf{X}_{a,t-1}^{v_m}, \beta_t\mathbf{I}\right) \tag{14}$$

where $\beta_t (t = 1, \ldots, T)$ is a variance schedule controlling the noise level at each step t. Here, the Gaussian noise added to the data after undergoing the entire forward diffusion process is formulated as follows:

$$q_\sigma\left(\mathbf{X}_{a,t}^{v_m} | \mathbf{X}_{a,0}^{v_m}\right) = \mathcal{N}\left(\mathbf{X}_{a,t}^{v_m};\ \sqrt{\overline{\alpha}_t}\mathbf{X}_{a,0}^{v_m}, (1-\overline{\alpha}_t)\mathbf{I}\right) \tag{15}$$

where $\bar{\alpha}_t = \prod_{i=0}^{t}(1-\beta_t)$. Furthermore, $\mathbf{X}_{a,t}^{v_m}$ can be directly obtained from $\mathbf{X}_{a,0}^{v_m}$ in closed form using the reparameterization trick

$$\mathbf{X}_{a,t}^{v_m} = \sqrt{\overline{\alpha}_t}\mathbf{X}_{a,0}^{v_m} + \sqrt{1-\overline{\alpha}_t}\boldsymbol{\epsilon},\ \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}) \tag{16}$$

where $\boldsymbol{\epsilon}$ is randomly sampled Gaussian noise. In this article, following the forward process, the proposed model generates noisy images of both modalities, denoted as $\mathbf{X}_{a,T}^{v_1}$ and $\mathbf{X}_{a,T}^{v_2}$.

The reverse process, which leverages the shared features of the target class as conditional inputs of the corresponding modality, is a critical process to generate trigger samples incorporated information of the target class from Gaussian noise. The reverse process involves progressively refining a noise-corrupted image to reconstruct the trigger samples with the target label of both modalities simultaneously over a fixed number of steps T. The conditional distribution of both modalities can be parameterized as follows:

$$p_\theta\left(\mathbf{X}_{a,0:T}^{v_1} \mid \mathbf{F}_c^{v_1}\right) = p\left(\mathbf{X}_{a,T}^{v_1}\right) \prod_{t=1}^{T} p_\theta\left(\mathbf{X}_{a,t-1}^{v_1} \mid \mathbf{X}_{a,t}^{v_1}, \mathbf{F}_c^{v_1}\right) \tag{17}$$

$$p_\theta\left(\mathbf{X}_{a,0:T}^{v_2} \mid \mathbf{F}_c^{v_2}\right) = p\left(\mathbf{X}_{a,T}^{v_2}\right) \prod_{t=1}^{T} p_\theta\left(\mathbf{X}_{a,t-1}^{v_2} \mid \mathbf{X}_{a,t}^{v_2}, \mathbf{F}_c^{v_2}\right). \tag{18}$$

With an extra class-specific shared feature of each modality, the conditioned sampling distribution of the corresponding modality is formulated as follows:

$$p_\theta\left(\mathbf{X}_{a,t-1}^{v_1} \mid \mathbf{X}_{a,t}^{v_1}, \mathbf{F}_c^{v_1}\right) = \mathcal{N}\left(\mathbf{X}_{a,t-1}^{v_1}; \boldsymbol{\mu}_\theta\left(\mathbf{X}_{a,t}^{v_1}, t, \mathbf{F}_c^{v_1}\right), \sigma_t^2\mathbf{I}\right) \tag{19}$$

$$p_\theta\left(\mathbf{X}_{a,t-1}^{v_2} \mid \mathbf{X}_{a,t}^{v_2}, \mathbf{F}_c^{v_2}\right) = \mathcal{N}\left(\mathbf{X}_{a,t-1}^{v_2}; \boldsymbol{\mu}_\theta\left(\mathbf{X}_{a,t}^{v_2}, t, \mathbf{F}_c^{v_2}\right), \sigma_t^2\mathbf{I}\right) \tag{20}$$

where

$$\boldsymbol{\mu}_\theta\left(\mathbf{X}_{a,t}^{v_m}, t, \mathbf{F}_c^{v_m}\right) = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{X}_{a,t}^{v_m} - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta\left(\mathbf{X}_{a,t}^{v_m}, t, \mathbf{F}_c^{v_m}\right)\right). \tag{21}$$

In the reverse process, a Mamba-based U-Net is designed to model the conditional Gaussian distribution $p_\theta(\mathbf{X}_{a,t-1}^{v_m} \mid \mathbf{X}_{a,t}^{v_m}, \mathbf{F}_c^{v_m}), m = 1, 2$, thus estimating the noise $\boldsymbol{\epsilon}_\theta(\mathbf{X}_{a,t}^{v_m}, t, \mathbf{F}_c^{v_m})$ added to $\mathbf{X}_{a,0}^{v_i}$. Then, the estimated noise is adopted to obtain $\mathbf{X}_{a,t-1}^{v_m}$

$$\mathbf{X}_{a,t-1}^{v_1} = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{X}_{a,t}^{v_1} - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta\left(\mathbf{X}_{a,t}^{v_1}, t, \mathbf{F}_c^{v_1}\right)\right) + \sigma_t\mathbf{z}^{v_1} \tag{22}$$

$$\mathbf{X}_{a,t-1}^{v_2} = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{X}_{a,t}^{v_2} - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta\left(\mathbf{X}_{a,t}^{v_2}, t, \mathbf{F}_c^{v_2}\right)\right) + \sigma_t\mathbf{z}^{v_2} \tag{23}$$

where $\mathbf{z}^{v_1}, \mathbf{z}^{v_2} \sim N(0, \mathbf{I})$, σ_t is continuous-time noise scale.

For each step of the reverse process, the denoising network predicts the noise components from noisy samples under the guidance of shared multimodal features. This requires the backbone to effectively propagate the shared semantic conditions across the feature representation, so that the generated trigger samples can preserve consistent target-class semantics and visually natural structures, which is particularly important for remote sensing images with large spatial extents and strong contextual correlations among land-cover regions. Therefore, Mamba is introduced into the backbone of U-Net to progressively generate the trigger sample of each modality. Benefiting from the four-directional scanning mechanism, Mamba can efficiently model long-range dependencies and contextual correlations during the iterative denoising process. In this way, the Mamba-based U-Net provides a suitable backbone for shared feature-conditioned diffusion trigger generation, enabling both semantic consistency and visual fidelity of the generated multimodal trigger samples. The Mamba-based U-Net consists of multiple stages of convolutional layers and VSS blocks. The output of each step is formulated as follows:

$$\boldsymbol{\epsilon}_\theta\left(\mathbf{X}_{a,t}^{v_1}, t, \mathbf{F}_c^{v_1}\right) = MambaU_t\left(Concat\left(\mathbf{X}_{a,t}^{v_1}, \mathbf{F}_c^{v_1}\right)\right) \tag{24}$$

$$\boldsymbol{\epsilon}_\theta\left(\mathbf{X}_{a,t}^{v_2}, t, \mathbf{F}_c^{v_2}\right) = MambaU_t\left(Concat\left(\mathbf{X}_{a,t}^{v_2}, \mathbf{F}_c^{v_2}\right)\right) \tag{25}$$

where $MambaU_t(\cdot)$ denotes the Mamba-based U-Net of tth step and $Concat(\cdot)$ is the concatenation operation. Finally, the generated trigger samples $\widehat{\mathbf{X}}_a^{v_1}$ and $\widehat{\mathbf{X}}_a^{v_2}$ can be obtained by reverse process with T steps.

To optimize the shared feature-conditioned Mamba-diffusion model, the trigger generation loss is designed, including cross-entropy loss and visual consistency loss. The trigger generation loss $\mathcal{L}_{trig}$ is defined as follows:

$$\mathcal{L}_{trig} = \alpha\mathcal{L}_{ce}\left(M_{\mathcal{O}}\left(\widehat{\mathbf{X}}_a^{v_1}, \widehat{\mathbf{X}}_a^{v_2}\right), y_c\right) + \beta\mathcal{L}_{consis}\left(\left(\mathbf{X}_a^{v_1}, \widehat{\mathbf{X}}_a^{v_1}\right), \left(\mathbf{X}_a^{v_2}, \widehat{\mathbf{X}}_a^{v_2}\right)\right) \tag{26}$$

where α and β are weighted coefficients of the cross-entropy term and visual consistency term, respectively. The cross-entropy loss $\mathcal{L}_{ce}$ is adopted to encourage the model to misclassify generated multimodal trigger samples into the target label y_c

$$\mathcal{L}_{ce}\left(M_{\mathcal{O}}\left(\widehat{\mathbf{X}}_a^{v_1}, \widehat{\mathbf{X}}_a^{v_2}\right), y_c\right)$$

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$$= -\frac{1}{A}\sum_{a=1}^{A}\left(y_c^a \log\left(M_{\mathcal{O}}\left(\widehat{\mathbf{X}}_a^{v_1}, \widehat{\mathbf{X}}_a^{v_2}\right)\right)\right) \tag{27}$$

where A is the number of trigger sample pairs. The visual consistency loss $\mathcal{L}_{consis}$ calculates the mean square error to guide the generated trigger samples to be close to the clean image for each modality

$$\mathcal{L}_{consis}\left(\left(\mathbf{X}_a^{v_1}, \widehat{\mathbf{X}}_a^{v_1}\right), \left(\mathbf{X}_a^{v_2}, \widehat{\mathbf{X}}_a^{v_2}\right)\right) = \sum_{b=1}^{B_1}\left\|\mathbf{X}_{a,b}^{v_1} - \widehat{\mathbf{X}}_{a,b}^{v_1}\right\|_2^2 + \sum_{b=1}^{B_2}\left\|\mathbf{X}_{a,b}^{v_2} - \widehat{\mathbf{X}}_{a,b}^{v_2}\right\|_2^2 \tag{28}$$

where B_1 and B_2 denote the number of bands of modality v_1 and v_2, respectively.

D. Multimodal Watermark Embedding

In the multimodal watermark embedding process, the special output of the multimodal remote sensing image classification model is defined when given the constructed multimodal trigger sample pairs. To embed the multimodal trigger sample pairs into the multimodal remote sensing image classification model as watermarking, the generated trigger samples are incorporated into the training phase alongside the clean training samples. The classification model is optimized using the cross-entropy loss to encourage the model to generate the predefined label

$$\mathcal{L}_{emb} = -\frac{1}{A}\sum_{a=1}^{A} y_p \log\left(\hat{y}_p\right). \tag{29}$$

In this way, the watermarked model can classify the multimodal trigger set into the predefined label successfully, while the model without watermarking returns incorrect predictions.

E. Ownership Verification

Once a model is suspected to be a stolen one, the owner can leverage the embedded watermark to verify the potential IP infringement. The proposed method adopts the black-box verification manner, which only requires querying the suspicious model through an API, without the need for internal details of the model.

Proposition 1: Let $\tilde{\mathbf{Y}}$ be the predicted labels of the suspicious model for the trigger set, and $\mathbf{Y}$ be the predefined labels of the trigger set. The predicted labels are compared with the predefined labels to calculate the classification accuracy of the suspicious model on the trigger set [referred to as watermark success rate (WSR)]. If the classification accuracy exceeds the threshold ϵ, the suspicious model is considered to contain a model watermark and can be identified as an unauthorized copy of the owner's model $M_{\mathcal{O}}^{\mathcal{W}}$.

IV. Experiment Results and Analysis

A. Experimental Setup

1) Datasets: The Trento dataset consists of an HSI and a LiDAR image collected over a rural area in Trento, Italy. The HSI contains 63 spectral bands, while the LiDAR image provides elevation information. The dataset has a spatial size of 600×166 pixels and includes six land-cover classes.

The Houston dataset was released by the 2013 GRSS Data Fusion Contest and contains an HSI and a LiDAR image acquired over the University of Houston and surrounding urban areas. The HSI includes 144 bands with a spatial resolution of 2.5 m. The dataset has 349×1905 pixels and contains 15 029 labeled samples from 15 land-cover classes.

The Augsburg dataset was collected in Augsburg, Germany, and consists of an HSI and a dual-polarimetric SAR image. The HSI contains 180 spectral bands, while the SAR image has four bands. Both images have a spatial size of 332×485 pixels, and the scene includes seven land-cover classes.

2) Models and Competing Methods: In the experiments, three widely used models, including SAL2RN [52], HCT [53], and IFN [54], are selected as target models to be protected to evaluate the watermarking methods. These models are based on different backbones, including CNN and Transformer, ensuring the diversity of the target models.

To demonstrate the effectiveness of the proposed method, several state-of-the-art black-box watermarking methods are adopted as competing methods, including ProtectIP [37], UHFB [55], MEA [41], and FunCouple [56]. Notably, the existing watermarking methods are designed to protect models with a single input, so we extend these methods to protect multimodal classification models for comparison.

3) Evaluation Metrics: For quantitative analysis, two widely used metrics, overall accuracy (OA) and Kappa coefficient (KC), are adopted to assess the accuracy of both clean and watermarked classification models. WSR, defined as the proportion of trigger samples classified into the assigned target labels, is used to validate the effectiveness of the watermarking methods. To evaluate the visual quality of the multimodal trigger samples, PSNR and SSIM are selected to compare the similarity between the trigger sample and the corresponding clean sample. In addition, we use SAM to further reflect the spectral difference between the trigger sample and the HSI sample.

B. Effectiveness and Fidelity

The effectiveness of watermarking methods is evaluated by their capability to accurately verify model ownership. Table I shows the effectiveness of competing methods and the proposed method across three models on three datasets. All methods demonstrate comparable effectiveness, showing WSR higher than 95%. On the Augsburg dataset, the WSR of most competing methods for the IFN model remains below 100%, with values mostly around 96%, which suggests a slight decrease in performance. WSR of the proposed method achieves 100% on all datasets. The results ensure the model owner can claim ownership with more confidence.

The watermarked models should ideally achieve classification accuracy comparable to clean models without watermarking. Therefore, the fidelity of watermarking methods is evaluated by comparing the classification performance of watermarked and clean models on three datasets. As shown in Table I, on the Trento dataset, the watermarked models

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TABLE I

FIDELITY AND EFFECTIVENESS OF DIFFERENT METHODS

SAL2RN	Trento			Houston			Augsburg		
	OA	**KC**	**WSR**	**OA**	**KC**	**WSR**	**OA**	**KC**	**WSR**
Clean	99.53%	99.37%	-	98.98%	98.86%	-	94.60%	92.18%	-
ProtectIP	99.76%	99.68%	100%	98.88%	98.79%	100%	93.86%	91.11%	100%
UHFB	99.58%	99.33%	100%	99.33%	99.28%	100%	95.35%	93.32%	99.33%
MEA	97.67%	96.90%	100%	98.16%	98.01%	100%	94.39%	91.83%	100%
FunCouple	99.46%	99.28%	100%	98.23%	98.08%	100%	93.95%	91.26%	100%
Proposed	99.76%	99.68%	100%	99.26%	99.20%	100%	94.87%	92.59%	100%
HCT	**OA**	**KC**	**WSR**	**OA**	**KC**	**WSR**	**OA**	**KC**	**WSR**
Clean	98.11%	97.46%	-	98.70%	98.59%	-	90.97%	87.19%	-
ProtectIP	96.63%	95.51%	100%	96.07%	95.74%	100%	89.89%	85.66%	100%
UHFB	98.15%	97.52%	100%	95.33%	94.94%	100%	87.45%	82.39%	100%
MEA	97.18%	96.23%	100%	96.50%	96.20%	100%	88.85%	84.25%	99.33%
FunCouple	98.12%	97.48%	100%	98.18%	98.02%	100%	85.04%	79.29%	99.33%
Proposed	98.36%	97.80%	100%	98.69%	98.58%	100%	90.75%	86.87%	100%
IFN	**OA**	**KC**	**WSR**	**OA**	**KC**	**WSR**	**OA**	**KC**	**WSR**
Clean	99.53%	99.37%	-	99.27%	99.21%	-	93.55%	90.85%	-
ProtectIP	99.59%	99.45%	99.33%	99.49%	99.45%	99.33%	92.77%	89.55%	96.00%
UHFB	99.64%	99.52%	100%	99.75%	99.73%	100%	94.94%	92.72%	96.67%
MEA	99.11%	98.82%	100%	98.90%	98.81%	100%	92.47%	89.37%	98.67%
FunCouple	99.41%	99.22%	95.33%	98.82%	98.72%	100%	91.94%	88.57%	96.67%
Proposed	99.47%	99.29%	100%	99.47%	99.43%	100%	95.01%	92.81%	100%

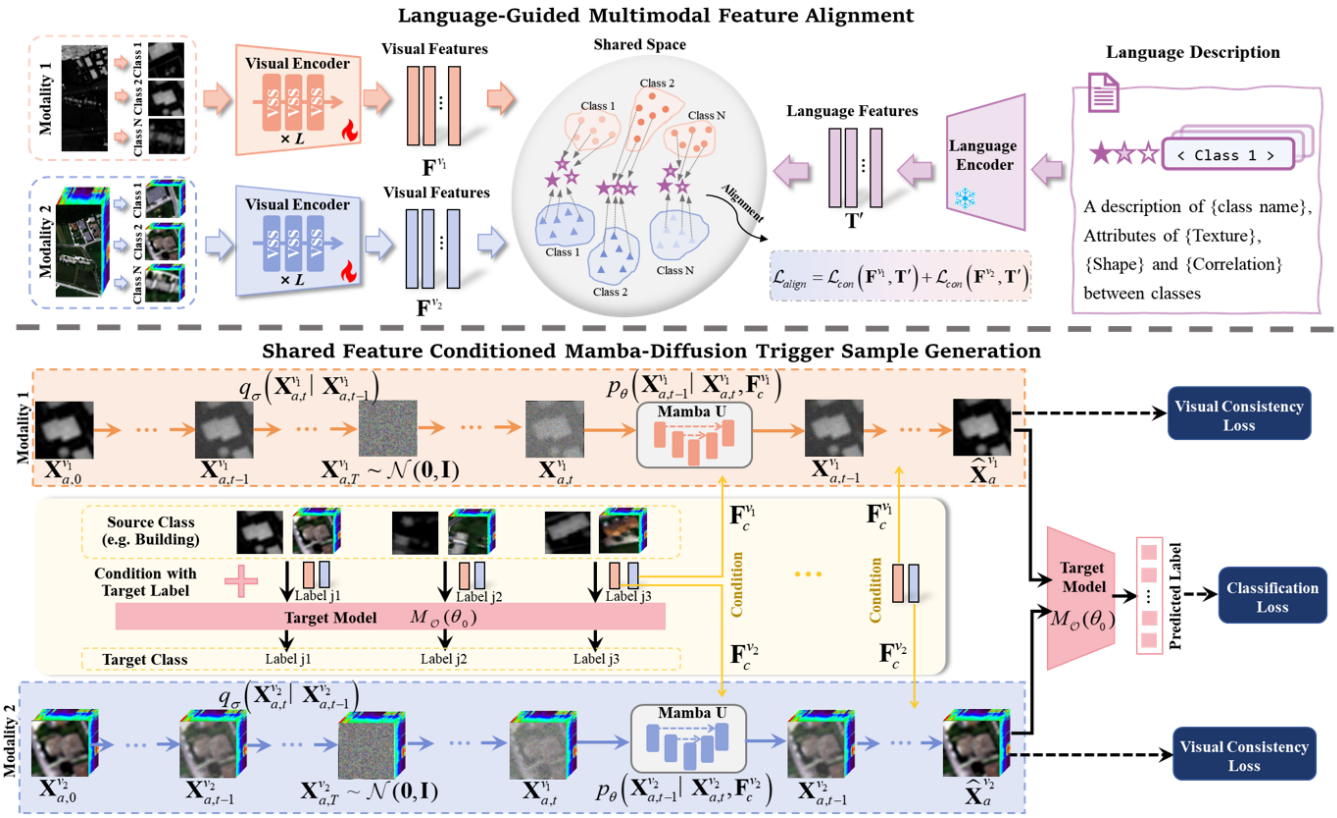


Fig. 4. Overview of language-guided multimodal collaborative trigger set generation method.

obtained using the MEA method exhibit a noticeable performance drop for both the SAL2RN and HCT models, with OA decreasing by nearly 2%. This is because the MEA generates trigger samples by combining source-class samples, which may introduce relatively strong perturbations into both modalities when extended to multimodal remote sensing data,

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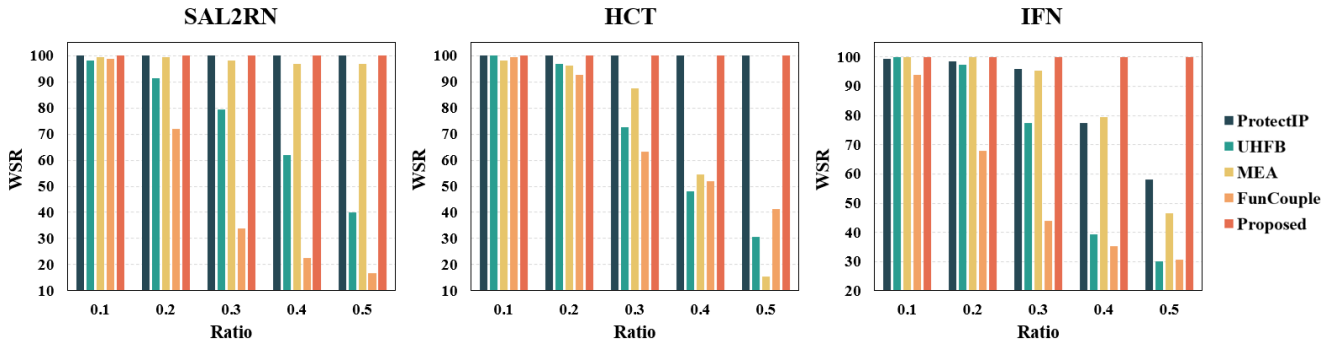


Fig. 5. Robustness of different methods against additive noise perturbations on three models.

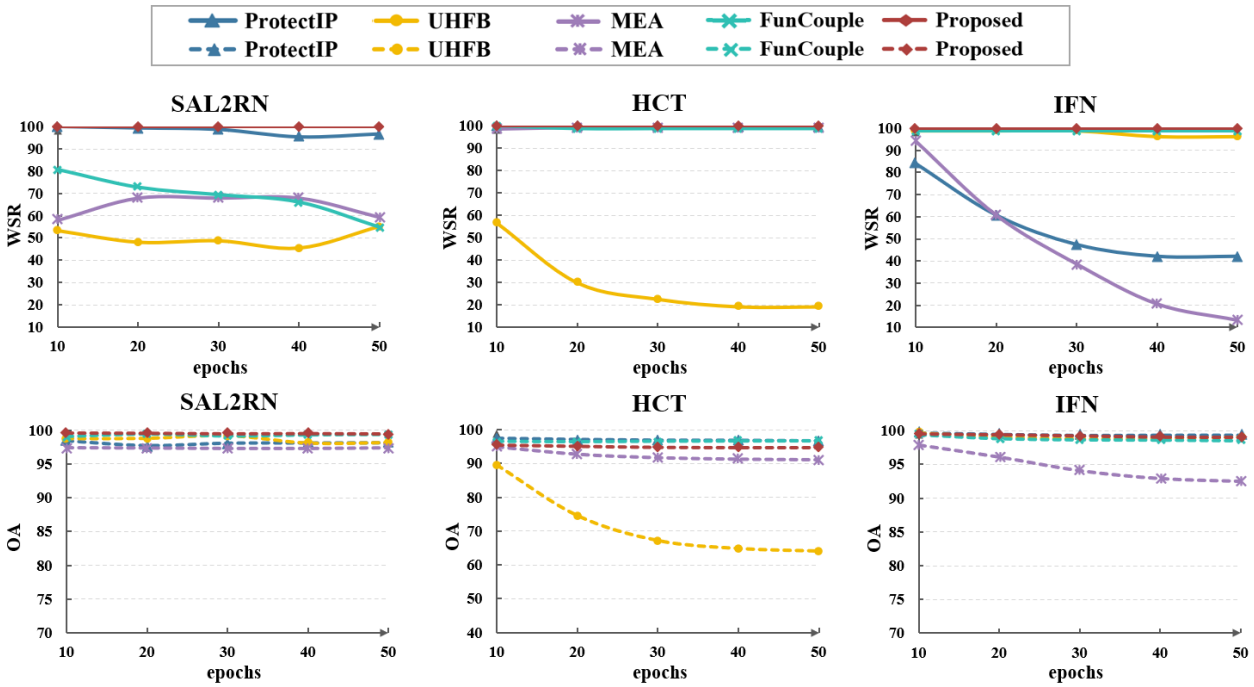


Fig. 6. Robustness of the different methods against model fine-tuning attack on three models.

thereby reducing clean classification accuracy. The FunCouple method exhibits relatively poor fidelity on the Augsburg dataset, especially for the HCT and IFN models. Notably, the KC of the HCT model drops by more than 7% after watermark embedding. The trigger samples of FunCouple are generated by convexly fusing two base samples from different source classes in the channel dimension. When extended to multimodal remote sensing data, especially complex scenes such as Augsburg, the fused samples may contain mixed land-cover semantics, making the model learn ambiguous watermark-related decision boundaries and thus degrading fidelity. The proposed method demonstrates superior fidelity, with OA degradation less than 0.5% on all datasets. This is because the proposed method generates language-guided multimodal collaborative triggers under shared semantic constraints, which better preserves the original multimodal decision capability after watermark embedding.

C. Robustness

1) Noise: Preprocessing input samples is a simple yet effective strategy for adversaries to reduce trigger distinguishability, thus mitigating model watermarking. A common approach is to add random noise, which may disrupt trigger-related patterns learned by the model. To evaluate robustness against noise attacks, random noise sampled from $\mathcal{N}(0, 0.1)$ is added to the input trigger samples through weighted averaging, with the noise ratio varying from 0.1 to 0.5. As shown in Fig. 5, UHFB and FunCouple suffer sharp drops, with WSR falling below 50% at a ratio of 0.5 for all models. Across all three models, the proposed method consistently maintains a 100% WSR, demonstrating strong robustness to noise-based attack.

2) Fine-Tuning: The fine-tuning attack assumes that the adversary only has access to a few samples in the test set and attempts to fine-tune the stolen model. For fine-tuning, we randomly sample data from the test set, with the number of samples set to 10% of the training set size, and train for a maximum of 50 epochs. Fig. 6 shows the OA and WSR of the watermarked model of different watermarking methods during fine-tuning. FunCouple shows superior robustness compared to other competing methods. ProtectIP also performs well on the SAL2RN and HCT models, with only minimal drops in OA and WSR. While MEA preserves classification accuracy, its

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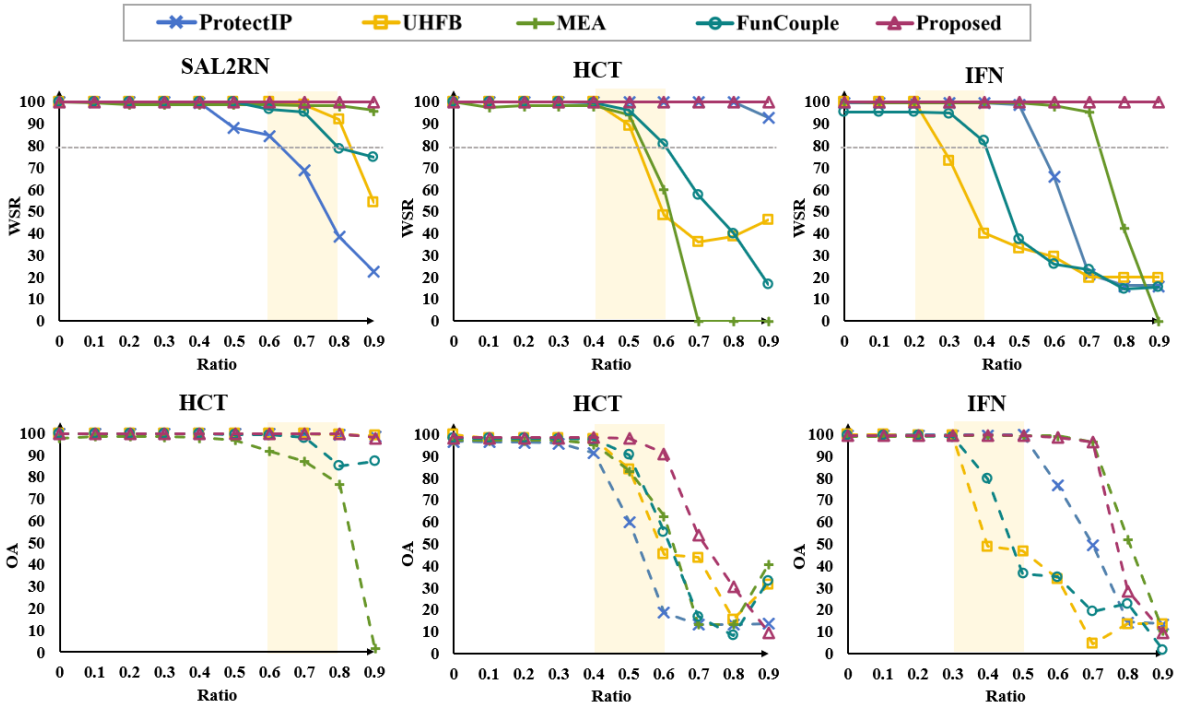


Fig. 7. Robustness of the different methods against model pruning attack on three models.

TABLE II
PERCEPTUAL STEALTHINESS OF DIFFERENT METHODS

Method	Trento			Houston			Augsburg		
	SSIM	PSNR	SAM	SSIM	PSNR	SAM	SSIM	PSNR	SAM
ProtectIP	0.5767	21.6927	7.7271	0.6668	28.5882	4.5218	0.6253	24.7211	6.1320
MEA	0.5995	22.6855	11.4478	0.7152	27.8714	10.9139	0.6334	25.7585	8.7868
FunCouple	0.9157	33.5870	5.0424	0.9196	36.1849	6.0462	0.8919	32.6297	6.5034
Proposed	**0.9848**	**40.4609**	**2.0817**	**0.9810**	**38.8478**	**2.6456**	**0.9982**	**49.8183**	**0.9591**

WSR drops significantly on SAL2RN and IFN, suggesting vulnerability to removal. The proposed method maintains WSR of 100% after 50 epochs of fine-tuning, which demonstrates the robustness of the proposed method against fine-tuning attack.

3) Pruning: To evaluate the robustness against model pruning attack, we adopt the l_1-norm unstructured pruning strategy with pruning rates ranging from 0% to 90%. Fig. 7 shows the classification accuracy and watermarking performance of different methods for different models. As the pruning rate increases, both metrics remain stable initially and degrade rapidly after exceeding a certain threshold. Compared to other methods, the proposed method still exhibits a high WSR with the pruning rate up to 90% for all models. It is worth noting that even when the models become ineffective on the classification task, the proposed method can still reliably verify the model ownership. The results indicate that the proposed method is robust to pruning attacks.

4) Overwriting: Watermark overwriting is a more advanced active attack that aims to weaken the original watermark while embedding a forged one into the same model. In the experiment, an independently constructed trigger set is used to implant an attacker-defined watermark into the watermarked models. OA, original WSR, and attacked WSR of different methods are reported in Fig. 8 to evaluate the classification accuracy and the effectiveness of the original watermark and forged watermark. The attacked WSR of all methods reaches 100%, indicating the forged watermark can be successfully embedded into the watermarked models. However, the original

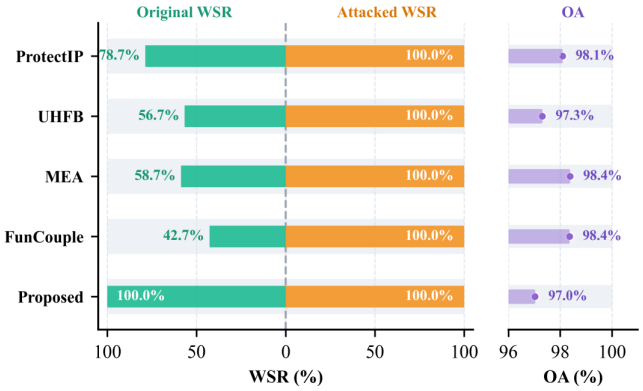


Fig. 8. Robustness of the different methods against model overwriting attack.

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TABLE III
PERFORMANCE AND OWNERSHIP VERIFICATION TIME OF THE PROPOSED METHOD WITH DIFFERENT NUMBERS OF TRIGGER SAMPLES

Number	SAL2RN				HCT				IFN			
	OA	KC	WSR	Time(s)	OA	KC	WSR	Time(s)	OA	KC	WSR	Time(s)
50	99.79%	99.72%	100%	0.05	98.53%	98.03%	100%	0.15	99.57%	99.43%	100%	1.23
150	99.76%	99.68%	100%	0.12	98.36%	97.80%	100%	0.20	99.47%	99.29%	100%	2.52
250	98.14%	97.51%	100%	0.18	99.79%	99.78%	100%	0.25	99.06%	98.74%	100%	3.63
350	98.70%	98.27%	100%	0.25	97.78%	97.03%	100%	0.27	98.94%	98.59%	100%	4.85
450	98.29%	97.71%	100%	0.29	97.23%	96.29%	100%	0.31	98.75%	98.33%	100%	5.90

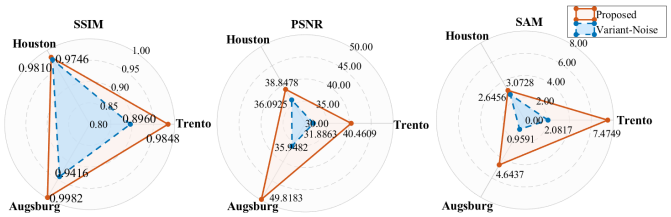


Fig. 9. Performance of the proposed trigger generation method and its variant using Gaussian noise as a condition (variant noise) on three datasets.

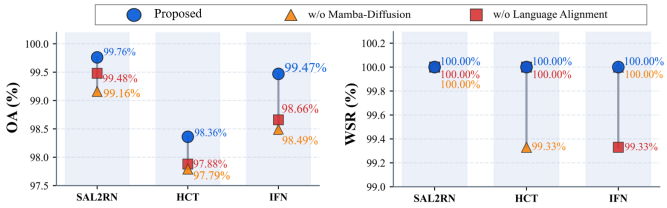


Fig. 10. Performance of the proposed LTDifMark and its variants without language alignment or replacing the Mamba-based U-Net with a conventional U-Net on three models.

WSR of the compared methods drops sharply, showing that the original watermark is severely weakened after overwriting. LTDifMark still maintains high original WSR while preserving an acceptable OA of 97.0%. This result indicates that although LTDifMark cannot prevent the insertion of a forged watermark, it can effectively retain the original multimodal watermark under overwriting attacks.

D. Perceptual Stealthiness

To evaluate the perceptual stealthiness of trigger samples, we employ SSIM, PSNR, and SAM as objective metrics. As shown in Table II, UHFB is excluded as its trigger samples are generated purely from random noise. The proposed method achieves consistently higher SSIM and PSNR across all datasets, indicating superior preservation of structural details and visual fidelity. Furthermore, in terms of SAM, ProtectIP yields better SAM on the Houston and Augsburg datasets compared to other competing methods, while the proposed method consistently surpasses ProtectIP on all datasets. The proposed method achieves SAM of 2.0817 on the Trento dataset and 0.9591 on the Augsburg dataset, demonstrating the capability to maintain spectral information.

E. Ablation Study

1) Effectiveness of Shared Feature Condition: The shared feature conditioned Mamba-diffusion trigger sample generation method is proposed to leverage class-specific multimodal shared features as a condition to guide the sample toward the target class, improving trigger stealthiness. To evaluate the effectiveness of shared feature conditioning, we conduct experiments that compare the performance of the proposed method and its variant using Gaussian noise as a condition (variant noise) on three datasets. As shown in Fig. 9, the trigger samples generated by variant noise exhibit lower image quality than those of the proposed method, with decreased SSIM and PSNR, and higher SAM values. The results demonstrate that the shared feature condition is essential for guiding the trigger samples toward the target class with fewer perturbations, balancing the collaborative trigger ability and imperceptibility.

2) Number of Trigger Samples: To investigate the impact of trigger sample number, we generate 50, 150, 250, 350, and 450 multimodal trigger samples for ownership verification. Table III reports the classification accuracy of watermarked models, WSR, and ownership verification time with different numbers of trigger samples on three datasets. As shown in Table III, WSR remains 100% under all settings, while the classification accuracy slightly decreases as the number of trigger samples increases. Therefore, 150 trigger samples are adopted to balance fidelity and effectiveness.

Since the generated trigger samples are retained as the query set after watermark embedding, ownership verification only requires querying the suspicious model with these samples and calculating the WSR, without additional trigger generation. As shown in Table III, the ownership verification time remains acceptable even with 450 trigger samples, demonstrating the practicality of the proposed method.

3) Effectiveness of Language Alignment and Mamba-Diffusion: To evaluate the effectiveness of the language-guided alignment and Mamba-diffusion trigger generation modules, we compare the proposed method with two variants: "w/o Language Alignment," which removes the language-guided multimodal feature alignment strategy, and "w/o Mamba-Diffusion," which replaces the Mamba-based U-Net denoiser with a conventional U-Net. As shown in Fig. 10, removing the language-guided alignment module leads to a consistent OA decline across the three models, reducing the OA from 98.36% to 97.88%, while the average WSR remains high. This degradation arises because the absence of alignment constraints makes modality-specific features inconsistent in the

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TABLE IV
QUANTITATIVE EVALUATION OF DIFFERENT TRIGGER CATEGORY DISTRIBUTIONS ON WATERMARK PERFORMANCE

Model	Trigger Category	OA	WSR
SAL2RN	Single Optimal	99.76%	100%
	Single Suboptimal	99.62%	100%
	Two Classes	98.93%	66.67%
HCT	Single Optimal	98.36%	100%
	Single Suboptimal	98.28%	100%
	Two Classes	98.70%	50%
IFN	Single Optimal	99.47%	100%
	Single Suboptimal	99.42%	100%
	Two Classes	98.08%	50%

shared semantic space, resulting in divergent trigger perturbations across modalities and affecting the original classification boundaries. Moreover, variant-w/o Mamba-diffusion yields the lowest OA on all datasets, although WSR remains relatively high. This decline is mainly because the weakened long-range dependence modeling and shared-condition propagation make the generated triggers less natural and more artifactual, introducing additional training noise during watermark embedding, thus affecting the performance on clean data.

4) Influence of Trigger Category Distribution: To evaluate the influence of trigger category distribution, we compare three settings with the same number of trigger samples, including a single optimal category (with highest PSNR), a single suboptimal category (with second-highest PSNR), and two categories. As shown in Table IV, both single-category settings achieve 100% WSR on the three models, indicating that trigger samples from different individual categories can be effectively embedded, while the optimal category provides slightly better OA. However, when triggers from two categories are jointly used, the WSR decreases significantly on SAL2RN, HCT, and IFN. This is because the current category-specific embedding rule requires the model to learn multiple trigger-label mappings simultaneously, where shared watermark perturbation patterns may dominate over category-specific semantic cues and cause different triggers to be mapped to the same response. Therefore, we adopt the single optimal category to ensure stable triggering and better model fidelity.

V. CONCLUSION

In this article, we propose a novel language-driven trigger-aligned diffusion method for multimodal remote sensing image classification model watermarking, enabling reliable IP protection. The main idea of this article is to fully consider the collaborative relationship of multimodal remote sensing data to guarantee the semantic consistency of the multimodal trigger sample, thereby improving the trigger ability of the model watermark. The language-guided multimodal collaborative trigger set generation is designed to align the class-specific shared features of different modalities to the same space with language priors, and the extracted multimodal shared features of the target class are adopted as a condition to guide the Mamba-diffusion model to generate multimodal trigger samples collaboratively. The designed strategy balances the relationship between collaborative trigger ability and stealthiness, making the watermark embedding stable and reliable. Extensive experiments are conducted to verify the effectiveness and fidelity of the proposed method compared to state-of-the-art model watermarking methods. Despite these promising results, several challenges merit future investigation. In practical remote sensing missions, data is often distributed across clients with varying modalities and heterogeneous model architectures. Developing watermarking methods adaptable to such federated learning frameworks is essential, yet remains complicated by severe modality heterogeneity. Future research will focus on developing federated watermarking strategies that are robust to these decentralized and heterogeneous constraints, ensuring reliable IP protection in real-world scenarios.

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